

Problem

A 240-V rms 60-Hz supply serves a load that is 10 kW (resistive), 15 kVAR (capacitive), and 22 kVAR (inductive). Find:

(a) the apparent power

(b) the current drawn from the supply

(c) the kVAR rating and capacitance required to improve the power factor to 0.96 lagging

(d) the current drawn from the supply under the new power-factor conditions

Step-by-step solution

Step 1 of 5

(a)

$$S = 10 - j15 + j22$$

$$S = 10 + j7 \text{ kVA}$$

$$S = |S| = \sqrt{10^2 + 7^2}$$

$$S = 12.21 \text{ kVA}$$

Step 2 of 5

(b)

$$S = VI$$

$$I = \frac{S}{V}$$

$$I = \frac{10000 + j7000}{240}$$

$$I = 41.667 - j29.167$$

$$I = 50.86 \angle -35^\circ \text{ A}$$

Step 3 of 5

(c)

$$Q_1 = \tan^{-1}\left(\frac{7}{10}\right)$$

$$Q_1 = 35^\circ$$

$$Q_2 = \cos^{-1}(0.96)$$

$$Q_2 = 16.26^\circ$$

$$Q_C = P_1[\tan \theta_1 - \tan \theta_2]$$

$$Q_C = 10[\tan 35^\circ - \tan 16.26^\circ]$$

$$\boxed{Q_C = 4.083 \text{ k VAR}}$$

$$C = \frac{Q_C}{\omega V_{rms}^2}$$

$$C = \frac{4083}{2\pi(60)(240)^2}$$

$$\boxed{C = 188.03 \mu\text{F}}$$

Step 4 of 5

(d)

$$S_2 = P_2 + j\theta_2$$

$$P_2 = P_1 = 10 \text{ k W}$$

$$Q_2 = Q_1 - Q_C$$

$$Q_2 = 7 - 4.083$$

$$Q_2 = 2.917 \text{ k VAR}$$

$$S_2 = 10 + j2.917 \text{ k VA}$$

Step 5 of 5

But,

$$S_2 = V I_2$$

$$I_2 = \frac{S_2}{V}$$

$$I_2 = \frac{10000 + j2917}{240}$$

$$I_2 = 41.667 - j12.154 \text{ A}$$

$$\boxed{I_2 = 43.4 \angle -16.28^\circ \text{ A}}$$